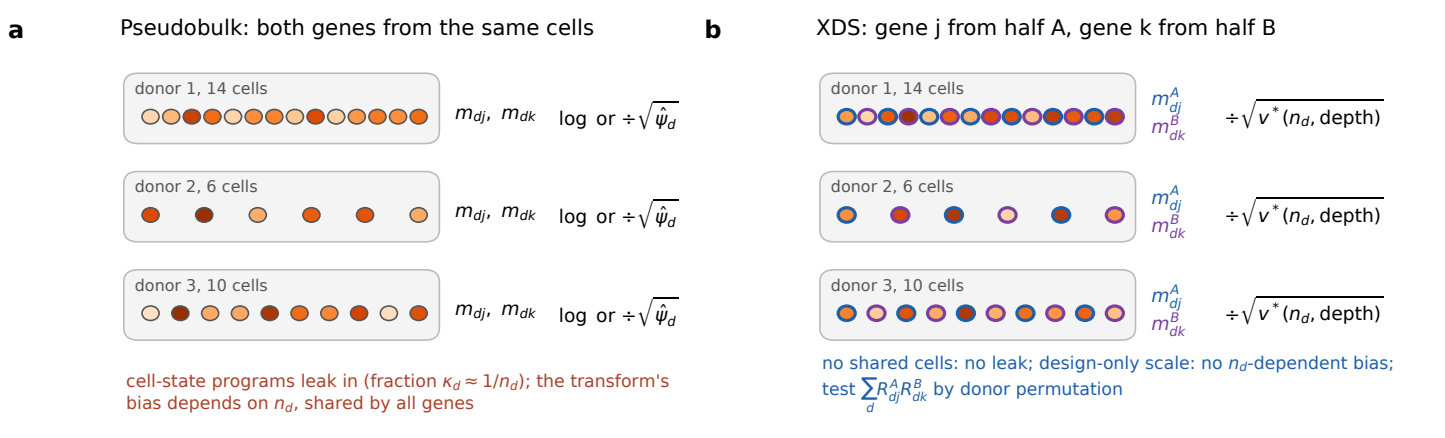
a
Pseudobulk: both genes from the same cells
donor 1, 14 cells
donor 2, 6 cells
donor 3, 10 cells
m_{dj}, m_{dk}
log or $\div\sqrt{\hat{\psi}_d}$
cell-state programs leak in (fraction $\kappa_d \approx 1/n_d$); the transform's bias depends on n_d, shared by all genes
b
XDS: gene j from half A, gene k from half B
donor 1, 14 cells
donor 2, 6 cells
donor 3, 10 cells
m^A_{dj}
m^B_{dk}
$\div\sqrt{v^*(n_d, \text{depth})}$
no shared cells: no leak; design-only scale: no n_d-dependent bias;
test $\sum_d R^A_{dj} R^B_{dk}$ by donor permutation